capturing the core facets of the research idea. A central challenge in this process involves obtaining a reproducible and comparable representation of research ideas that contains all the information necessary for novelty understanding and judging. To this end, we adapt existing research idea templates (Si et al., 2025; Shahid et al., 2025) to structured presentations consisting of the following aspects:

- **Problem statement:** A detailed description of the core research problem(s) or question(s) addressed.
- **Objective:** A clear articulation of the research aim(s) or intended outcomes.
- **Solution approach:** A detailed description of the proposed methods or approaches designed to solve the problem and achieve the stated objectives.

Following this, we synthesize the individual reviewer justifications for their novelty scores into a single, coherent justification aligned with each gold-standard novelty judgment score. To achieve this, we provide an LLM with the reviewers' textual comments, the generated research idea, the assigned gold novelty score, and the novelty judgment rubric as context. The model is then prompted to identify the reviewers' arguments justifying their given novelty scores for a research idea and to integrate them into a unified, coherent justification that explains the rationale behind the gold novelty score.

Finally, after involving an LLM in earlier stages of data processing and accounting for their tendency to produce hallucinations and inaccuracies, our final step focuses on two objectives: enriching research ideas with relevant related works to enable grounded novelty judgments, and enforcing strict quality control to ensure that only high-quality samples are included in the final dataset. To this end, we first obtain related works by retrieving the titles and abstracts of publications cited in the introduction and related work sections of paper submissions. We extract the relevant paper sections from the PDF submissions using Nougat (Blecher et al., 2024) and obtain the works cited therein via Semantic Scholar (Kinney et al., 2025). Citations in other sections are ignored, as they typically include references to evaluation metrics or datasets that are irrelevant for novelty assessment. Next, we apply a quality filtering step. Research ideas are excluded if fewer than five related works are retrieved, typically due to missing indexing in Semantic Scholar. We then use an LLM to verify the formal correctness of the research idea, ensuring it is written in the first person, not as a summary of multiple reviews, and contains no explicit numerical novelty scores. Additionally, we assess whether all arguments in the synthesized novelty justifications are fully grounded in the corresponding research ideas and related works. Ungrounded justifications may arise not only from LLM-induced hallucinations during the synthesis of reviewer arguments, but also when reviewers use arguments derived from related works that are not cited in the corresponding submitted paper, or when related works are not available via Semantic Scholar. To assess the grounding of these justifications, we use an LLM to verify that every argument in the textual novelty justification is grounded in either the research idea or in the retrieved related works. In particular, the LLM extracts all arguments from the novelty justification pertaining to the novelty aspects of the research idea and verifies whether each novelty argument is grounded in the idea. In addition, the LLM extracts all arguments involving comparisons to related work and ensures that each argument is grounded in at least one retrieved title and abstract. Only samples with a formally correct research idea and a fully grounded novelty justification are included in the final data set.

4.1.2. Dataset

Our final dataset consists of 1,381 research ideas, each paired with rubric-based novelty scores, corresponding textual novelty judgment justifications, and an average of 25.23 titles and abstracts of related works. We perform a stratified 80:20 train-test split on the dataset, yielding the data distribution presented in Table 2.

Novelty Score	**#training**	**#test**	$\sum$
1	60	15	75
2	239	60	299
3	349	87	436
4	322	81	403
5	134	34	168
$\sum$	1,104	277	1,381

Table 2: Class distributions within RINoBench.

4.2. Evaluation Metrics

This section outlines the metrics used in RINoBench to evaluate both the predicted novelty scores and the generated textual justifications for the novelty judgments.

4.2.1. Novelty Score Metrics

$\mathbf{F_1}$ We treat novelty score prediction as a classification task and use F_1 as the primary evaluation metric. Specifically, we employ macro-averaged F_1 to ensure that each degree of novelty is weighted equally. This approach disregards the actual category imbalance and treats all categories as equally important, consistent with how they are valued in practice. Additionally, we report the F_1 scores for each novelty category individually to evaluate how a model performs across different categories, identifying where it performs particularly well or poorly.

Mean Absolute Error (MAE) Since novelty scores in this task are not limited to discrete categories but also represent rubric-based values, MAE allows us to evaluate the magnitude of deviation between predicted and gold scores. This provides insight not only into whether correct novelty scores are predicted, but also into how far the predictions diverge from the gold standard. By evaluating the average distance of predictions from the gold scores, we can determine if a model's predictions are reasonably close or significantly misaligned with the expected outcomes.

4.2.2. Justification Metrics

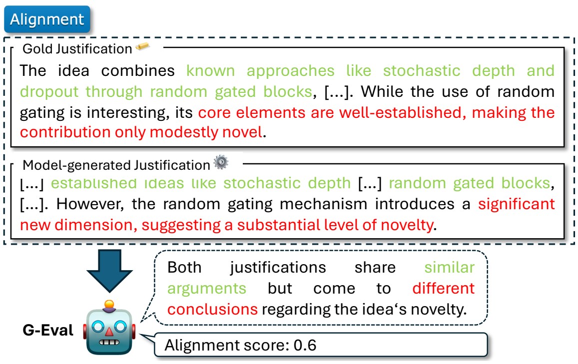

Figure 2: Evaluation of justification *alignment* for novelty judgments using the G-Eval framework, which produces textual reasoning and a numerical score. We use only the numerical score for evaluation.

Alignment Evaluating the alignment of novelty judgment justifications is essential to ensure that a model's decision-making process mirrors human-like judgment, both in terms of logic and argumentation. This metric evaluates whether the reasoning behind a predicted novelty judgment is consistent with the reasoning in the gold standard. Specifically, it verifies whether a model-generated justification follows the same line of argumentation, presents similar supporting arguments, and reaches the same conclusion as the human gold justification. As illustrated in Figure 2, we utilize the G-Eval framework (Liu et al., 2023) to prompt an LLM for alignment evaluation, generating an alignment score that ranges from 0 (worst) to 1 (best).

Known Aspects Recall We measure the extent to which arguments in the gold novelty justification that pertain to "*known aspects*" (see Figure 3) are

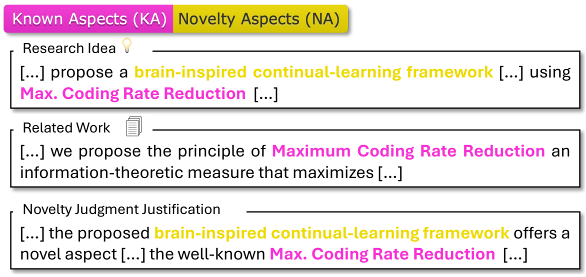

Figure 3: Example illustrating *known* and *novelty* aspects in novelty judgment justifications. *Known aspects* refer to elements in a justification that highlight already established concepts or findings from previous work in a research idea. *Novelty aspects* denote elements in a justification that highlight new contributions of a research idea, which do not exist in prior work.

captured in a model-generated justification. Following the FActScore approach (Min et al., 2023), an LLM first extracts all arguments from both the model-generated and gold justifications. It then verifies whether each argument extracted from the model-generated justification is supported by the gold justification. The final metric is computed as:

$$\text{Recall}_{\text{Args.}} = \begin{cases} \min\left(1, \frac{N_{\text{supp}}}{N_{\text{gold}}}\right) & \text{if } N_{\text{gold}} > 0, \\ 0 & \text{if } N_{\text{gold}} = 0. \end{cases} \tag{1}$$

where N_{supp} is the number of model-generated known-aspect arguments supported by the gold justification, and N_{gold} is the total number of known-aspect arguments in the gold justification. Figure 4 provides an illustrated example.

Novelty Aspects Recall Analogous to above, we measure the extent to which arguments in the gold novelty justification related to "*novelty aspects*" (see Figure 3) are captured by a model-generated justification. This is done using the same LLM-based argument extraction and validation approach as in *Known Aspects Recall*, with the metric computed using Equation 1. In this case, N_{supp} represents the number of model-generated novelty-aspect arguments supported by the gold justification, while N_{gold} denotes the total number of novelty-aspect arguments in the gold justification.

Additional Known Aspects Ratio We measure the extent to which a model generates additional known-aspect arguments, which are not present in the gold justification but grounded in the associated related works. To evaluate this, we again use the same LLM-based argument extraction and validation approach as in *Known Aspects Recall*. Additionally, the LLM verifies whether the known-aspect

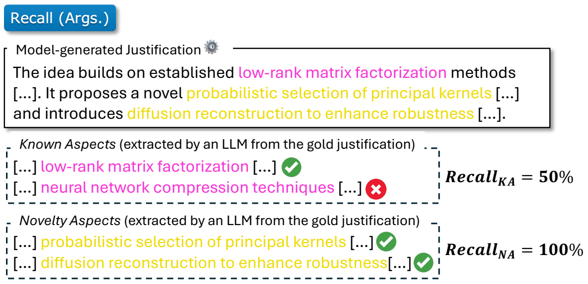

Figure 4: Example of *Known Aspects Recall* and *Novelty Aspects Recall* for evaluation of novelty judgment justifications.

arguments extracted from the model-generated justification are grounded in the corresponding related works. The metric is then computed as:

$$\text{Ratio}_{\text{Additional}} = \frac{N_{\text{additional}}}{\max(N_{\text{gold}}, 1)} \quad (2)$$

where $N_{\text{additional}}$ is the number of model-generated known-aspect arguments unsupported by the gold justification but grounded in the related works and N_{gold} is the total number of known-aspect arguments in the gold justification. Figure 5 provides an illustrated example.

Additional Novelty Aspects Ratio Similarly to above, we assess the extent to which a model generates additional novelty-aspect arguments that are not present in the gold justification but are grounded in the respective research idea. This is achieved using the same LLM-based argument extraction and validation approach as in *Known Aspects Recall*, with an added LLM-based step to verify whether the extracted novelty-aspect arguments from the model-generated justification are grounded in the corresponding research idea. The metric is computed using Equation 2, where $N_{\text{additional}}$ is the number of model-generated novelty-aspect arguments unsupported by the gold justification but grounded in the research idea and N_{gold} is the total number of novelty-aspect arguments in the gold justification.

Known Aspects Hallucination Rate We quantify the extent to which model-generated justifications contain hallucinated known-aspect arguments (i.e., those not supported by any of the corresponding related works). To this end, we adopt the same LLM-based argument extraction and validation approach used in *Known Aspects Recall* and compute the metric as:

$$\text{Hall. Rate} = \begin{cases} \frac{N_{\text{hallucinated}}}{N_{\text{generated}}} & \text{if } N_{\text{generated}} > 0, \\ 0 & \text{if } N_{\text{generated}} = 0. \end{cases} \quad (3)$$

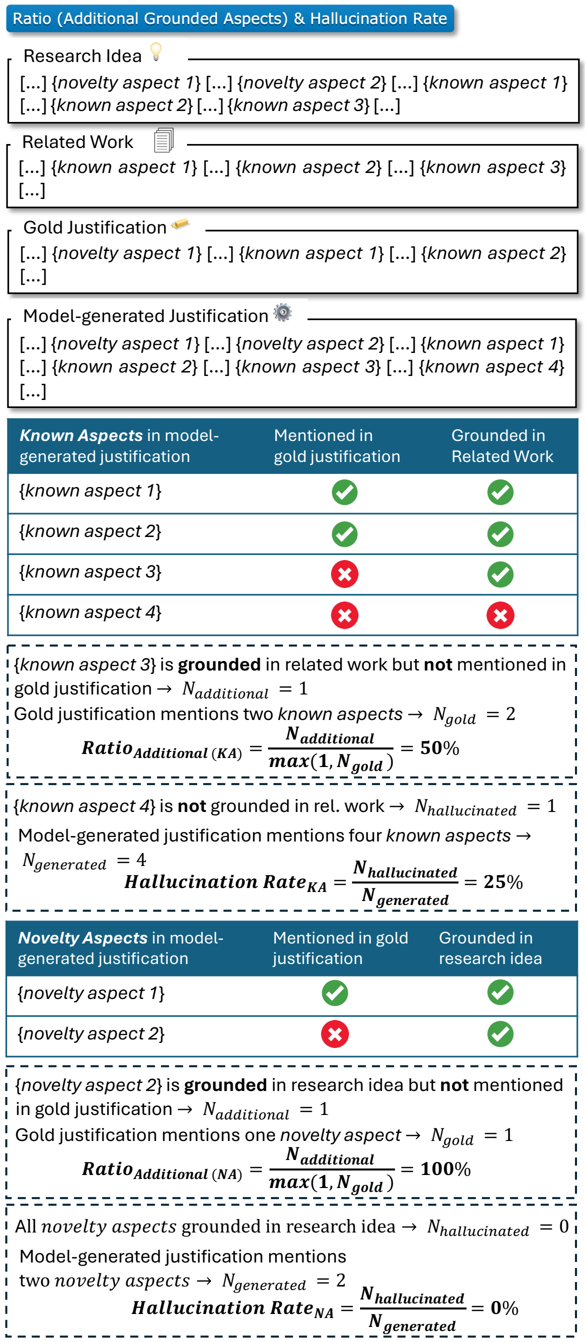

Figure 5: Example evaluation of a model-generated novelty judgment justification using *Additional Ratio* and *Hallucination Rate* for *known aspects* and *novelty aspects* respectively.

where $N_{\text{hallucinated}}$ is the number of model-generated known-aspect arguments unsupported by any of the corresponding related works and $N_{\text{generated}}$ is the total number of known-aspect arguments in the model-generated justification. Figure 5 provides an illustrated example.

Novelty Aspects Hallucination Rate Similarly, we quantify the extent to which model-generated